



**Maker Sprint-
STEAM Resource Package for the Busy Teacher**

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Biography

I am a teacher from Singapore with about 15 years of teaching experience. I began my career teaching biology and chemistry at a junior college, and today, I teach science to younger children in a primary school. My teaching journey across different educational levels has deepened my passion for improving scientific literacy and inspiring students to apply what they learn in science to drive change.

Beyond my classroom duties, I serve as a head of department at Chongfu School, where I oversee science education and its Applied Learning Programme. In this role, I lead curriculum design, pedagogical innovation, and assessment strategies, while also driving interdisciplinary collaboration and student-centred learning. A key part of my work involves facilitating professional development and fostering a culture of continuous learning among our teachers.

Alexandra K. Trenfor once said, “The best teachers are those who show you where to look but don’t tell you what to see.” This quote resonates with me. I firmly believe in empowering students to think critically and construct their own understanding, rather than confining their learning to prescribed answers. Thus, I actively champion teaching approaches that promote active learning, curiosity, and discovery. I am always eager to explore new ways to level up how science is being taught —making it relevant and engaging by connecting concepts to everyday life through hands-on activities, play, maker projects, and game-based learning.

While the curriculum often offers limited flexibility, I believe teachers have the freedom and should be intentional in how we deliver content in the classroom. Teaching, to me, is a shared journey. I look forward to collaborating with fellow educators to redesign learning experiences and make learning meaningful and enjoyable for our students.

Executive Summary

This project focuses on empowering teachers to facilitate active science learning experiences that incorporate principles from Science, Technology, Engineering, Arts and Mathematics (STEAM) education, maker education, and design thinking (DT). It aligns closely with the Singapore Science Curriculum Framework, which emphasises providing students with opportunities to inquire like scientists and to innovate using science.

In active learning, students engage in meaningful tasks such as questioning, discussing, investigating, modelling, or explaining. These tasks require them to apply scientific ideas to solve problems and mirror the authentic practices of scientists - building, testing, and refining ideas. Through these hands-on activities, students develop critical, adaptive, and inventive thinking while deepening their conceptual understanding of science.

However, teachers worldwide juggle demanding workloads, making it challenging to consistently incorporate innovative practices in their classrooms. To support teachers in realising Singapore's vision for science education, I am developing "Maker Sprint: a STEAM resource package." This practical toolkit is designed to complement existing primary science resources by offering maker-centric lesson ideas that enable teachers to efficiently design high quality and engaging active learning experiences.

The resource and instructional package consists of ready-to-use, adaptable lesson ideas inspired by classrooms and makerspaces in both Singapore and the United States. It comprises three key components. Drawing inspiration from the "project on a page" concept, *Spark and Create- Lesson on a Page*- is crafted with time-strapped teachers in mind. Each lesson plan provides a concise overview of a maker-centred lesson, with options for differentiated instruction, reflective questioning, and inclusive practices. *Idea Sprint* offers a quick glance through a collection of maker-centred lesson ideas sorted by levels and topics,

while *Maker Playground* introduces adaptable maker activities that could be carried out in a designated room or makerspace with specialised tools outside of the core curriculum.

At its core, this project embraces the view of science as an evidence-based, model-building discipline that seeks to understand and explain the real world. The strategic use of maker activities is grounded in the belief that joy and curiosity are catalysts for deep learning. Model-making not only sparks interest and motivation but also creates meaningful contexts for discovery and exploration. By supporting time-strapped teachers with this practical toolkit, the project aims to promote active learning in science classrooms, nurture resilience in problem-solving, and ultimately foster scientific literacy and a lifelong passion for science in students.

Overview of the Topic

Science is an evidence-based, model-building enterprise to understand the real world. In active science learning, students engage in meaningful tasks, such as questioning, discussing, investigating, modelling, or explaining, which require them to apply scientific ideas to solve problems. These experiences mirror the nature of science through practices such as building, testing, and refining models while cultivating critical, adaptive, and inventive thinking. By embedding principles derived from Science, Technology, Engineering, Arts and Mathematics (STEAM) education, maker education, and design thinking (DT) into the design of science learning experiences, learning becomes more engaging and meaningful, thereby promoting deep learning.

The strategic use of maker activities is grounded in the belief that joy and curiosity are catalysts for deep learning. Model-making not only sparks interest and motivation but also creates meaningful contexts for discovery and exploration. The making and use of physical and visual models in science education can support students in making sense of complex ideas and strengthens their conceptual understanding and systems thinking. When combined with playful learning and interdisciplinary problem-solving, model-making can further foster scientific literacy, collaborative skills, and the ability to connect knowledge across subjects.

Primary observations made in the Singapore and U.S. classrooms indicate that teachers often lament having limited time to plan for innovative classroom practices and quality lessons. To support the science teachers in realising the visions and objectives of science education, I initiated the development of the “Maker Sprint: a STEAM resource package.”

It is a practical toolkit designed to complement existing resources available to teachers in the implementation of the primary science syllabus. It includes ready-to-use,

adaptable maker-centric lesson ideas inspired by classrooms and makerspaces in both Singapore and the United States. Drawing inspiration from the concepts of “project on a page”, the lesson plans—titled *Spark and Create- Lesson on a Page*—are crafted with time-strapped teachers in mind. Each plan provides a concise overview of a maker-centred lesson, along with options for differentiated instruction, reflective questioning, and inclusive practices that promote active participation. The package also includes *Idea Sprint*, which is a table that offers a quick glance through a collection of primary science maker-centred lesson ideas sorted by levels and topics. *Maker Playground* is another collection of adaptable maker-centred activities that could be carried out in a designated room or makerspace with specialised tools outside of the core curriculum. The key feature of this STEAM resource package is that it enables teachers to quickly obtain ideas and rapidly design lessons that integrate STEAM, maker, and DT principles with playful learning. It enables teachers to design active learning experiences more efficiently while maintaining instructional quality and engagement.

By supporting time-strapped teachers with this practical toolkit, the project aims to promote active learning in science classrooms and nurture scientific literacy, resilience in problem-solving, and ultimately, a lifelong passion for science in students. This project aligns closely with Singapore’s national priorities in STEAM education and the development of emerging 21st-century competencies and skills in our students. It supports efforts to prepare students for future academic and career success in a rapidly evolving landscape.

Background and Primary Observations

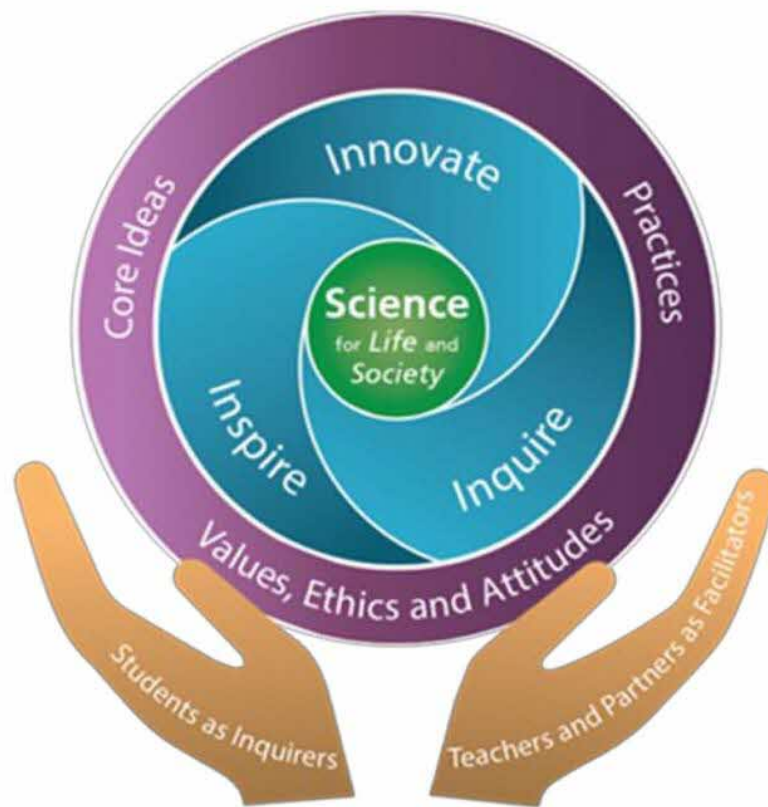
To me, science education should be a fun and hands-on adventure where we explore, discover and connect science to everyday lives through active inquiry and experiences, which can take place within and outside the physical classrooms. As DeRosa and Abruscato (2019a) said, one should not proceed to learn science by approaching it as a body of knowledge to be learnt. That would be equivalent to skipping a movie and watching only the final scene, and the ending would not make sense and be meaningful if one does not know the plot. The process of science learning should be an active one, as it is an endeavour that seeks knowledge in a systematic way. In active learning, students engage in meaningful tasks—such as questioning, discussing, investigating, modelling, or explaining. These tasks require them to use and apply scientific ideas to solve problems.

My personal philosophy on the teaching and learning of science is closely aligned with the principles set forth in the Singapore Science Curriculum Framework (Figure 1). Science education in Singapore has the dual goals of nurturing scientific literacy and providing strong science fundamentals for students, as well as building the nation's pipeline of innovators for future learning and careers in Science, Technology, Engineering and Mathematics (STEM) (Ministry of Education Singapore, 2024). “Inspire, Inquire and Innovate”—these three words encapsulate Singapore's vision for the overall experience of our students in science education. In addition, the collaboration in learning and teaching extends beyond students and teachers and includes partners who could support and facilitate students' understanding of the relevance of science to daily life, societal issues, and environmental contexts.

Figure 1

Science Curriculum Framework

(Ministry of Education Singapore, 2024)



The STEM curriculum in Singapore is one that places emphasis on hands-on and applied interdisciplinary learning. According to the Trends in International Mathematics and Science Study (TIMSS) 2023, conducted by the International Association for the Evaluation of Educational Achievement (IEA), Singapore's Primary 4 and Secondary 2 students continue to demonstrate strong competencies in Mathematics and Science, including foundational numeracy, scientific literacy, problem-solving as well as scientific-inquiry skills (Fishbein et al, 2025). These findings affirmed that students in Singapore are benefiting from the holistic STEM curriculum put in place and indicate that our teachers have adopted effective teaching approaches to support student learning in STEM subjects.

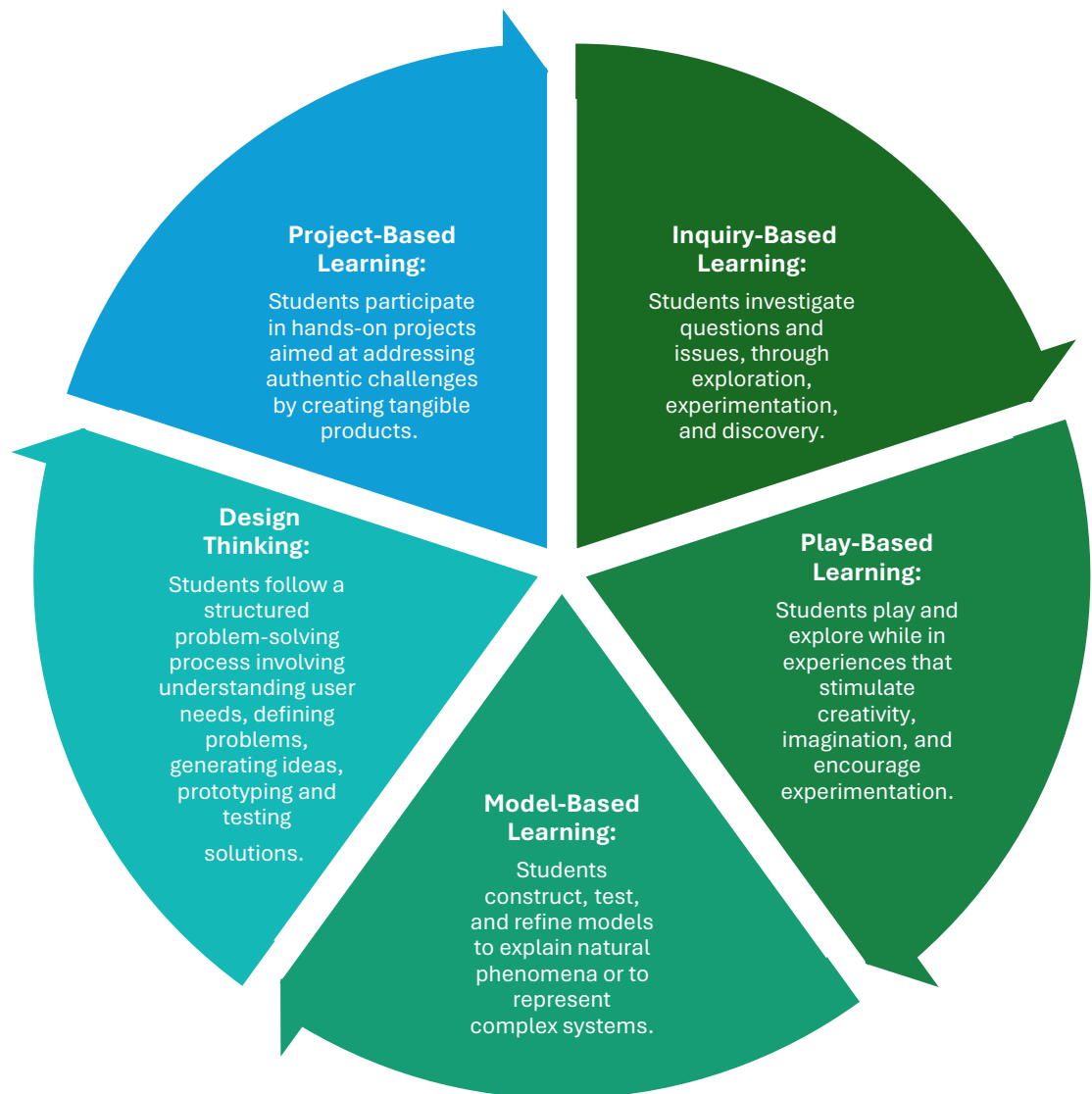
Maker Education

Since the 2010s, maker education has been recognised as a transformative way for STEM teaching and learning (Blikstein, 2013). It is a relatively new educational paradigm that focuses on delivering a constructivist, project-based, hands-on learning curriculum and instruction experience to students (Gerstein, 2019). In essence, it involves students learning by doing—they could be building, creating, and solving problems through self-directed projects. Since children need to develop both gross motor abilities and fine motor skills, well-planned science inquiry-based experiences through making activities can support this development (DeRosa & Abruscato, 2019b).

Project-based, inquiry-based, play-based, model-based learning as well as design thinking (DT) are some teaching approaches commonly identified to be effective for STEM instruction (Figure 2). These pedagogical approaches also underpin the active, iterative, and creative environments encouraged in makerspaces and maker-centred classrooms. In these approaches, we shift beyond rote memorisation of facts toward learning science through active sense-making, modelling, and problem-solving. Furthermore, at the heart of these approaches is their ability to empower students to discover solutions independently. Such meaningful learning not only deeply impacts classroom engagement, but it also has the potential to transform students' perceptions of what they can achieve (National Inventors Hall of Fame, 2020).

Figure 2

Teaching Pedagogies Commonly Identified to be Effective for STEM Instruction.



Infusing maker education into science teaching and learning is beneficial because it extends the core strengths of discovery-based science learning—such as observation, inquiry, and experimentation—into real-world contexts. When engaged in maker practices, students move beyond simply acquiring scientific knowledge and skills. They apply these competencies creatively through design, construction, and problem-solving activities that mirror authentic scientific practices—they will go through what real scientists and engineers

do in their jobs to solve real-world problems. This active engagement not only strengthens personal ownership of learning but also forges interdisciplinary connections (Gerstein, 2019).

Maker projects frequently fall short of perfect results on the first try, and this fosters resilience and adaptability among learners. The iterative process helps students acquire key skills valuable to succeed in a fast-evolving world. Moreover, understanding technology is increasingly critical in today's digital era. Exposure to advanced tools and programmes like artificial intelligence (AI), 3D printing and coding programmes in makerspaces allows students to enhance their technological proficiencies. There is also evidence that engaging in maker activities positively influenced students' self-efficacy beliefs in STEM and interest towards STEM-related careers (Chen et al., 2024; Forbes et al., 2021).

Science, Technology, Engineering, Art and Mathematics (STEAM) Education

Including Arts in STEM transforms it into STEAM, emphasising creativity, design, and human expression alongside STEM. This combination enriches learning by merging artistic approaches with technical skills, resulting in a deeper understanding and innovative solutions. Think of STEAM as a vibrant tapestry of interconnected subjects woven together with creativity. The inclusion of the arts enriches STEM learning, making educational experiences more engaging. Whether using STEM or STEAM approaches, it is the holistic nature of these experiences that most benefits the students. Together, STEAM and maker education help students integrate knowledge across disciplines and apply it in practical, creative ways.

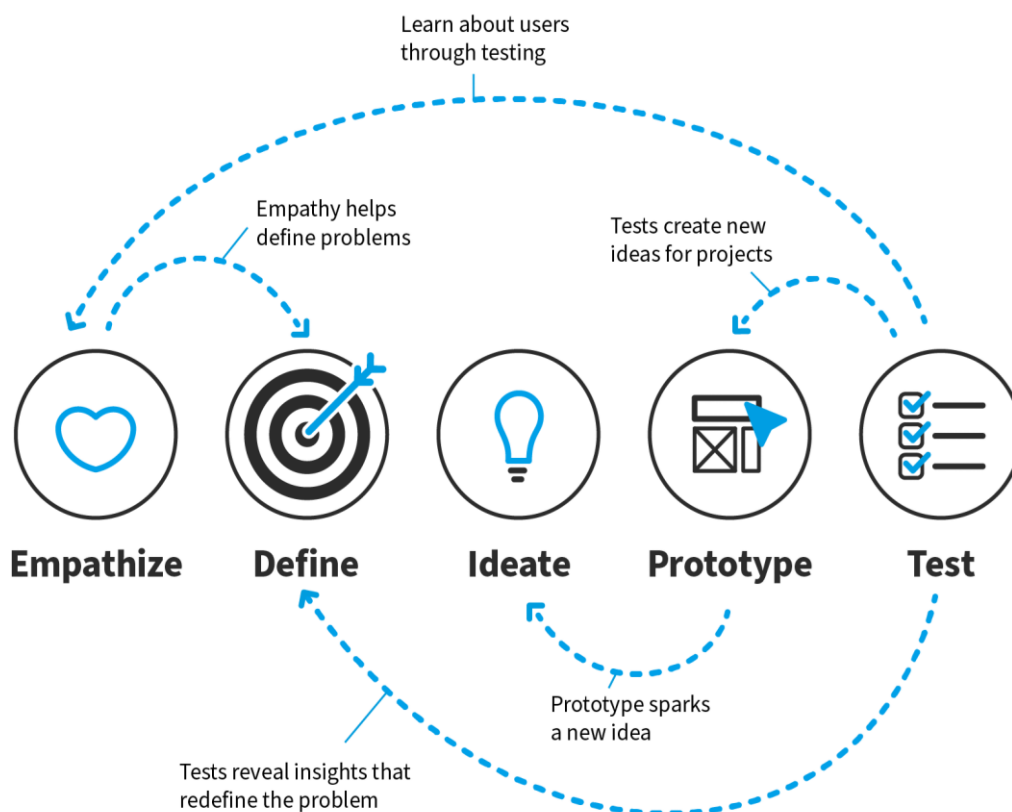
Design Thinking (DT)

Design thinking(DT) is a non-linear, iterative and human-centric process that enables students to understand the end-user needs, challenge existing perspectives, redefine problems, and develop creative solutions through prototyping and testing (Interaction Design Foundation, 2016). It is useful for addressing complex and ambiguous challenges and consists of five key stages: Empathise, Define, Ideate, Prototype, and Test (Figure 3).

Figure 3

Design Thinking Consists of Five Stages and is a Non-Linear Process to Approach Complex Problems.

(Interaction Design Foundation, 2016)



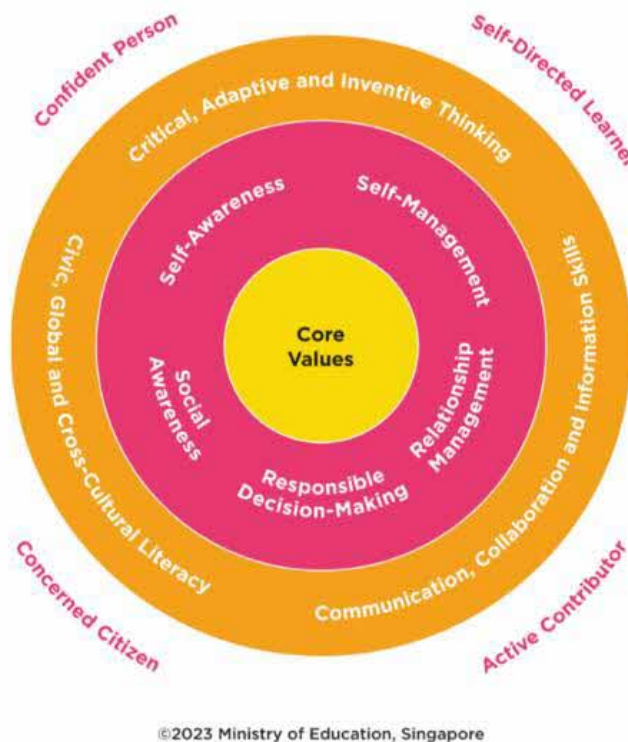
In itself as an approach used to solve problems, DT is effective for use in STEM education in K-12 classrooms to enhance students' understanding of STEM, develop critical thinking skills, and foster the development of a mindset open to ambiguity and experimentation (Vieira, 2024). Additionally, progressing through the five phases of DT during problem-solving facilitates the conduct of formative and peer assessments at both the individual and group levels. In this digital age, when assignments could be created at a click with artificial intelligence (AI), DT supports the critical transition from summative to formative assessment of learning and skill mastery. The use of the DT approach can allow teachers to more accurately and appropriately assess critical thinking skills and authenticate students' work continuously over a period.

Taken together, STEAM education, maker education and the DT approach to problem solving hold potential for fostering the emerging 21st-century competencies and skills essential for our students to be successful in navigating a future which is Brittle, Anxious, Nonlinear, and Incomprehensible (BANI). In Singapore, the Ministry of Education has identified these competencies and skills to be critical, adaptive, and inventive thinking; communication, collaboration and information skills; as well as civic, global, and cross-cultural literacy (Figure 4).

Figure 4

Singapore's Ministry of Education– Framework for 21st Century Competencies and Student Outcomes

(Ministry of Education Singapore, 2023)



Overview of Science, STEAM and Maker Education in Singapore

In Singapore's primary schools, our work in designing science learning experiences is well-supported by a generic integrated suite of curriculum resources, which is made available to all public primary schools. These resources are updated every 8 to 10 years and the most recent update was rolled out and implemented in schools in 2023. The resources include textbooks, activity books, *Science Pack – Activity Resource Kits for LEarning* (SPARKLE) and curated online lessons and videos on the Student Learning Space (SLS)¹. The syllabus

¹ <https://www.learning.moe.edu.sg/about/overview-of-student-learning-space/>

document² that accompanies this suite of curriculum resources spells out aims and guiding principles for recommended teaching strategies and assessment practices (Ministry of Education Singapore, 2024). These documents and resources are crafted and curated by a committee of experienced science teachers and academics in the local education fraternity (Nanyang Technological University Singapore, 2025).

Teachers attend professional development sessions to understand the intent behind the design of and how to use the resources. They are also encouraged to use their professional judgement instead of adhering strictly to the syllabus during implementation, to prioritise the interests and capabilities of the students. Teachers are advised to diversify their teaching methods and integrate resources and ideas from multiple sources to enrich science learning (Figure 5).

Figure 5

Science Instructional Strategies That Could Be Used in Science Classrooms

(as suggested in Singapore's Primary Science Syllabus 2023)



² https://www.moe.gov.sg/-/media/files/primary/syllabus/primary-science-syllabus-2023_may24.pdf

Maker education in Singapore schools is defined to comprise both analogue and digital making activities. Analogue making activities refer to hands-on creation with physical materials and tools (e.g., sewing and woodworking), while digital making activities involve students utilising technology and digital tools for design, creation, and innovation (e.g., 3D printing, 3D modelling and coding). Makerspaces and Design & Technology working studios are set up in some schools to provide students with access to a resource-rich environment and support to create and make. Efforts are also put in to cultivate a community of innovators and a collaborative network within and between schools. Through initiatives such as the STEM Applied Learning Programme³, many schools actively embed maker-centred activities to help students bridge abstract academic concepts with tangible understanding through practical application to solve real-world challenges. In Singapore, maker education is not only well-supported but is also recognised as a vital approach for enriching education by offering student-centred and experiential opportunities that foster creative, adaptive thinking and robust problem-solving skills.

Pennsylvania– STEELS K-12 Standards and Science Teacher Preparation

The K-12 Science, Technology & Engineering, and Environmental Literacy & Sustainability (STEELS) standards in Pennsylvania were adopted in January 2022 by the State Board of Education. These new academic standards focus on incorporating a multidimensional and integrated approach to STEM learning and prioritise a shift away from rote learning of scientific facts toward increasing students' involvement in sustained investigations to support deep understanding (Commonwealth of Pennsylvania- Department of Education, 2022).

³ <https://www.moe.gov.sg/secondary/schools-offering-full-sbb/school-specific-opportunities/applied-learning-programme>

To support the rollout of the STEELS standards, there has been significant time spent on discussion of the STEELS standards and great focus placed on discovery learning, hands-on activities, and fostering curiosity through questioning and exploration during the science teacher preparatory courses at the Indiana University of Pennsylvania (IUP). Instructors actively engage student teachers in discussions on practical lesson ideas and guide them to enact these effectively in K-12 science classrooms. For instance, some courses feature student teachers taking turns to lead their peers—acting as young learners—through “learning by doing” experiences. Resources such as the online STEELS Hub and training sessions further support in-service teachers in implementing these standards. This shift in teacher preparation and curriculum highlights the broader educational trend in the U.S., where hands-on learning and maker education are increasingly embedded within the K-12 curriculum.

Pennsylvania Schools- STEAM and Maker Education

In Pennsylvania elementary schools, science instruction typically ranges from 75 to 240 minutes per week, varying by grade level and school district. While STEAM-themed instruction can be part of the regular curriculum, many schools also run STEM (or STEAM) specials as designated time blocks during the school week, just like specials for art, music, library, or physical education. At Indiana’s Eisenhower Elementary School, fourth and fifth graders receive a dedicated weekly 40-minute STEAM lesson. Similarly, at Pittsburgh Brookline PreK-8, there is a weekly 45-minute STEM lesson for students from grades 1-8. During these specials, students develop baseline technological skills, computational thinking skills and apply scientific principles through making and engineering projects.

Maker culture is also widely observed in many aspects of life in the U.S., extending beyond schools. Libraries, museums and county fairs often include simple maker stations and corners for children to tinker, fostering early engagement in creative exploration. Moreover,

adults embody the maker and entrepreneurial spirit through popular practices such as do-it-yourself (DIY) crafts and activities, upcycling and thrifting, reflecting a vibrant culture of resourcefulness and creativity. Together, these aspects confirm a great emphasis on experiential, maker-centred learning as a foundational component of modern science education in the U.S.

The Busy Teacher

While STEAM, maker education and DT approaches offer transformative opportunities for student learning, their successful implementation hinges on sufficient logistical support as well as thoughtful planning and preparation by teachers. In their paper, Chen et al. (2024) classified the constraints and issues for making integration in formal classroom settings as pedagogical, the logistical and structural aspects of schooling (Table 1).

Table 1

Constraints and Issues in Making Integration into Formal Educational Contexts
(Chen et al., 2024)

Pedagogical	Logistical	Structural
• Teaching approach	• Funding	• Learning standards
• Teacher expertise	• Dedicated space	• Assessment of learning
	• Scheduling	

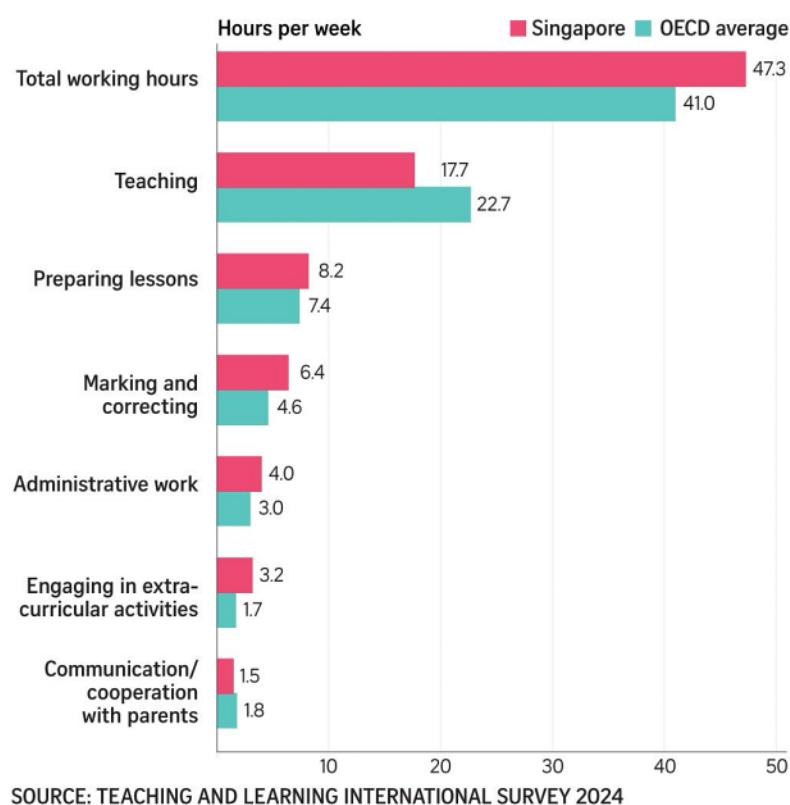
Teachers across cultures have different working hours, and how they spend their working hours also varies widely by country. Based on the findings of the Teaching and Learning International Survey (TALIS) 2024, full-time U.S. teachers in lower secondary schools report working hours at 46 hours per week. As cited in Tushara (2025), full-time

Singapore teachers reported longer working hours at 47.3 hours per week, as compared to the OECD average of 41 hours. In addition, Singapore teachers reported spending 8.2 hours per week on preparing lessons (higher than the OECD average: 7.4), and this has increased by 0.9 hours since 2018 (Figure 6). In the U.S., full-time teachers in lower secondary schools typically spend around 6 to 8 hours per week on lesson preparation and planning.

Figure 6

A Comparison between a Singaporean Teacher and the OECD Average of the Working Hours Spent Weekly on Various Tasks in Schools

(Tushara, 2025)



Delivering high-quality science and maker-centred lessons demands significant time for careful planning and lesson design. However, teachers around the world manage demanding schedules and extensive work duties. This adds challenges to incorporating

innovative teaching practices into the classroom. Furthermore, Ramli and Talib (2017) found that many teachers lack a thorough understanding of how to effectively integrate STEM into lessons, on top of other challenges including curriculum constraints, insufficient training, scarce resources and student engagement issues. To help teachers create meaningful learning experiences that truly benefit students, targeted professional development, providing sufficient resources, and ongoing support are essential.

Educator Project Implementation

Implementing making as an educational paradigm in science classrooms presents several challenges. This project aims to answer the following research question:

How might we encourage teachers to implement active science learning experiences without substantially increasing their preparation workload?

To address this, a resource and instructional package was specifically designed and developed for busy science teachers with limited time to plan maker-centred STEAM lessons.

This resource and instructional package (*a working draft of which is included in the appendices of this report*) provides lesson ideas and plans that integrate STEAM, maker education, and DT, complementing existing integrated curriculum materials in both countries. It fosters interdisciplinary connections and supports the development of 21st century skills and competencies in students. The format balances practical, ready-to-use resources with flexibility, enabling teachers to adapt lessons to their unique contexts. To ensure scalability, the resources will be piloted in one school, then shared and refined for broader use beyond the initial department or institution.

With access to this resource and instructional package, teachers will be better equipped to efficiently plan engaging maker-centred lessons that promote key scientific skills such as modelling, hands-on problem-solving, and systems thinking. The project also aims to deepen

teachers' and students' understanding of science as an evidence-based, model-building discipline by highlighting practices such as building, testing, and refining models framed within the DT process. Appendices A, B, and C include infographics summarising STEAM education, Design Thinking, and Maker education. These infographics give teachers who are unfamiliar with these approaches, a quick introduction to each teaching approach.

Ultimately, this project seeks to cultivate a positive attitude toward science learning by encouraging playful, meaningful, and creative experiences—nurturing curiosity, critical thinking, adaptability, and a passion for lifelong learning among both teachers and students.

Spark and Create—Lesson on a Page

In their book, Clark A. Campbell and Mick Campbell (2013) demonstrate how the complexity of a project can be distilled into a concise, one-page document using the One-Page Project Manager (OPPM) template, thereby facilitating clear and effective communication with stakeholders. In developing maker-centred lesson plans for busy teachers within this resource and instructional package, my primary objective is to minimise the cognitive load for teachers associated with planning, while preserving the crucial pedagogical and creative benefits for students. By adapting some principles from the OPPM, I strive to make the lesson plan clear, concise, and easy to read within minutes.

My lesson plan template, titled *Spark and Create—Lesson on a Page*, is crafted with time-strapped teachers in mind. It capitalises on the principal strength of the OPPM concept by distilling from the inherent complex lesson details to clearly highlight the main components of the lesson. The focus is on presenting clear and concise lesson details at a glance to help the busy teacher quickly decide if the lesson is suitable and feasible for implementation. The method aims to convey substantial information swiftly, accurately and with clarity. The template sections are clearly marked out and organised to highlight

feasibility indicators upfront (Figure 7). It provides a concise overview of a maker-infused lesson, along with possibilities for differentiated instruction, reflective questioning, and inclusive practices that promote active participation among students. Appendix D includes sample one-paged lesson plans of two distinct maker-centred lesson.

Figure 7

Template for Spark and Create- Lesson on a page

SPARK AND CREATE- LESSON ON A PAGE:		LEVEL& SUBJECT:
LEARNING OUTCOME(S):	TOTAL DURATION:	PRODUCT EXAMPLE(S): [INSERT PHOTO]
INTERDISCIPLINARY LINK(S):	SKILL(S): SPECIALISED SKILL(S):	
INTRODUCTION / CONTEXT:		REFERENCE(S):
		SAFETY PRECAUTION(S):
MATERIALS:	DESIGN THINKING STEPS- TO FOCUS ON: ☐ Empathise: ☐ Define: ☐ Ideate: ☐ Prototype: ☐ Test:	PREPARATION: ACTIVITY STEPS:
DIFFERENTIATION: CHALLENGE FOR DEEPER INQUIRY (C) / EXTENSION FOR BROADER LEARNING (E):		DIFFERENTIATION: MODIFICATION FOR LOW READINESS OR MORE SUPPORT (M)
REFLECTION QUESTIONS:		ASSESSMENT/SUCCESS CRITERIA: ☐ ☐
REAL-LIFE APPLICATIONS:		

It is important to emphasise that these one-page lesson plans are not intended to replace the comprehensive lesson plans typically developed by teachers, particularly those who are new to the profession. Rather, they serve a specific purpose: to offer teachers a concise overview of the essential considerations before implementing a preparation-intensive, maker-centred lesson. Depending on their level of expertise and teaching experience, teachers should still seek to customise the lesson and undertake the necessary detailed planning and preparation as competent professionals.

Idea Sprint- Collection of Ideas for Primary Science Maker Activities

In addition, there is a supplementary table that serves as an archive of diverse, adaptable and ready-to-use lesson ideas that could be used in the science classroom. The primary objective of this compilation is to broaden teachers' pedagogical repertoire by exposing them to innovative, hands-on learning possibilities that actively engage students in scientific inquiry and experimentation. By sharing these varied examples, teachers are better equipped to facilitate experiential learning opportunities that promote critical thinking, creativity, and deeper conceptual understanding during science lessons. As shown in the template in Figure 8, the table is kept simple with the lesson ideas sorted by level and topic. It only includes brief information on the science concept(s) involved and task instructions.

Figure 8*Idea Sprint Collection of Primary Science Maker Activities*

Idea Sprint— Collection of Primary Science Maker Activities			
Level/Topic	Science Concept / Skill	Task	Product to be made
P1 & P2			[INSERT PHOTO]
P3			[INSERT PHOTO]
P4			[INSERT PHOTO]
P5			[INSERT PHOTO]
P6			[INSERT PHOTO]

Appendix E shows a working draft of *Idea Sprint*. The lesson ideas were compiled from classroom observations in both the United States and Singapore, as well as from a literature review of current educational practices. The collection is organised by age appropriateness and categorised according to the topics outlined in Singapore’s 2023 Primary Science Syllabus. This structure allows science teachers to easily access and explore a wide range of maker-centred activities tailored to different developmental stages and curricular goals.

Maker Playground—Collection of Maker Activities (Outside of Curriculum)

Maker Playground is a collection of a diverse range of adaptable maker activities that can be conducted outside of curriculum hours, such as during recess or after school in a designated room or makerspace. These activities may involve the use of craft resources and specialised tools, offering students additional opportunities for creative exploration and hands-on learning beyond regular class time. The collection of ideas serves as a resource for teachers to support flexible implementation and enrich the maker experience outside the classroom. As shown in the template in Figure 9, the table is kept simple and only includes brief information

on theme, task instructions, materials needed. Appendix F shows a working draft of *Maker Playground*.

Figure 9

Maker Playground— Collection of Maker-centred Activities outside of Curriculum

Maker Playground— Collection of Activities (outside curriculum)			
Theme	Task / Learning goal(s)	Tools Needed	Prototype
			[INSERT PHOTO]

Inclusion of Pictures of Prototype

Besides being clear and concise, an intentional design feature for *Spark and Create-Lesson on a Page*, *Idea Sprint*, and *Maker Playground* is the consistent inclusion of pictures depicting prototypes for each lesson and activity idea presented. This visual strategy is rooted in the adage that “a picture paints a thousand words.” The pictures effectively communicate the deliverable more clearly than text alone.

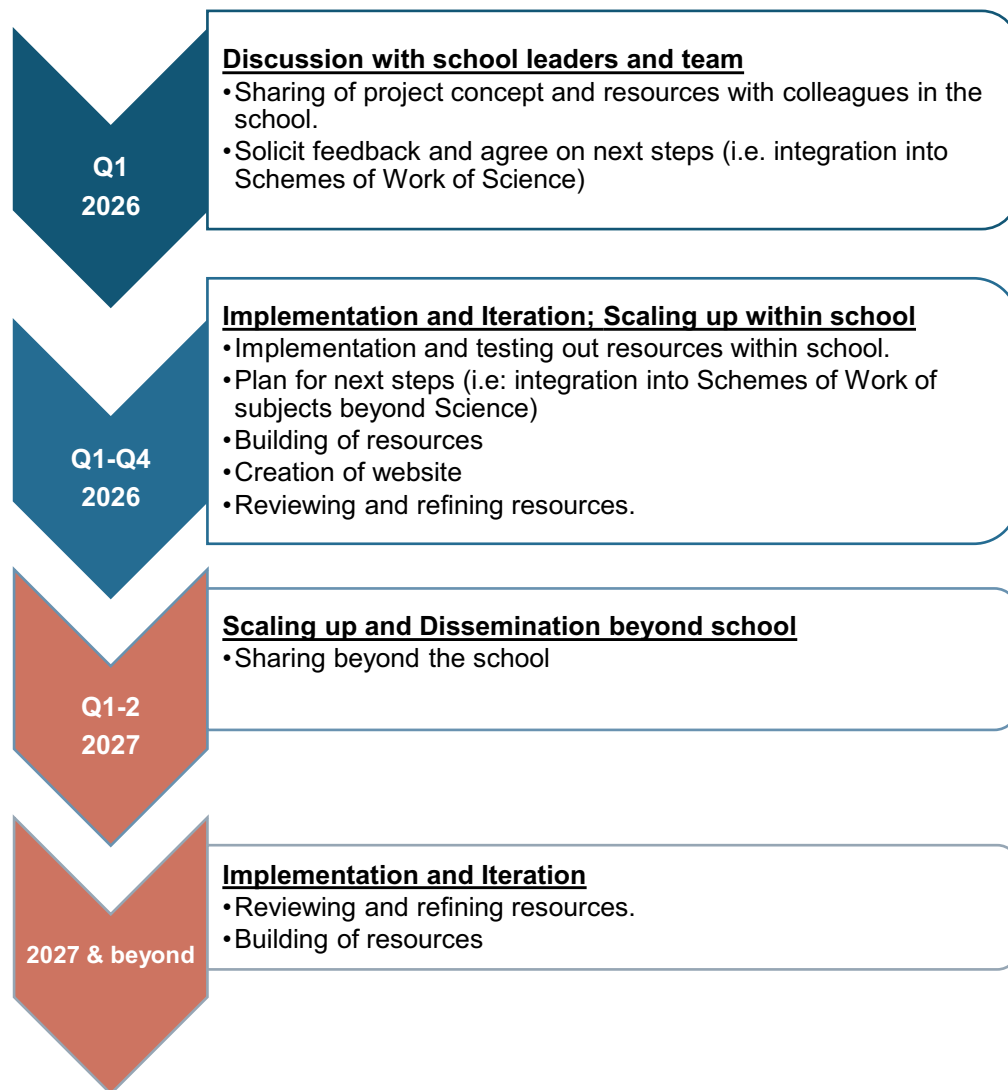
For the primary audience of these resources—teachers—the inclusion of prototype pictures serves several functions. Firstly, it demystifies the implementation process by visually demonstrating that the activities are both feasible and accessible within typical classroom settings. Secondly, they play a crucial role in bolstering teachers’ confidence, reassuring them that they can successfully guide their students toward creating comparable products. The intentional use of these pictures not only improves comprehension of the ideas but also facilitates teacher empowerment.

Timeline

My team and I have initiated the promotion of DT, STEAM and maker education within my school since 2024. I intend to continue driving this piece of work upon return to Singapore. I plan to begin by incorporating at least one maker-centred lesson in science classes for each level from Primary 3 to Primary 6. Where appropriate and meaningful, we will consider collaborating with other subject teachers to involve students in interdisciplinary projects. I am also deliberating and support the use of this resource and instructional package with a series of short, bite-sized professional development sessions. This could be done in a blended approach, where teachers can learn asynchronously online.

The instructional package could also be presented on a dedicated maker education website, possibly featuring downloadable resources, classroom video examples, and teacher testimonials to enhance accessibility for busy educators, while facilitating collaboration and resource sharing among schools worldwide.

For gathering feedback and incorporating reflective practice for teachers' professional reflection, a Google Form has been crafted (screenshots of the form are included in the Appendix G). Teachers will be invited to complete the form after they have tried out activities in the resource and instructional package. After collation of feedback and conduct of reviews, I intend to share these resources with more teachers through the professional learning communities beyond the school. My project timeline is summarised in Figure 10.

Figure 10*Project Timeline for Maker Sprint- STEAM Resource Package for the Busy Teacher***Anticipated Roadblocks**

One challenge is in securing buy-in from teachers who may already have packed schedules and varying familiarity with STEAM education, maker education and DT approaches. Thus, they might hesitate to invest additional time learning new methods or integrating hands-on maker activities into their lessons. They could also have varying levels of competency in science content mastery and maker skills, and this could create anxiety.

There could also be challenges in finding teachers across different subjects to collaborate on a multidisciplinary project.

In addressing this challenge, we would need to look into providing differentiated professional development opportunities in a bite-sized manner to build confidence and obtain buy-in. Teachers need to have a mindset shift. During these sessions, we should seek to highlight success stories and evidence of student engagement. We could also consider peer collaboration opportunities for teachers to share experiences and learn from one another. This would allow us to build a supportive professional learning community within the school.

Another challenge we would face would be the tight schedules and extensive curricular demands that could restrict the flexibility teachers have to introduce maker-centred activities. These activities often require longer blocks of time for exploration, iteration, and reflection, which can conflict with rigid period structures and standardised content coverage. Thus, there could be difficulty in integrating these activities meaningfully within existing time frames.

For this, there are modular and flexible maker activities that can be adapted to shorter or longer time frames based on teachers' schedules. Furthermore, we could work to integrate maker projects within mandated curriculum topics to ensure they reinforce required standards rather than competing with them. Finally, we could also explore the use of blended or flipped classroom practices to maximise time efficiency and enhance the learning experience. This way, more class time could be used for discussions and hands-on activities.

Lastly, traditional assessment practices emphasising factual recall and standardised testing may not align well with the hands-on, process-oriented learning central to maker education. This tension could discourage teachers from fully adopting maker approaches unless assessment systems evolve to value and capture diverse forms of student learning and

growth. In addressing this, we could develop formative assessment tools and rubrics aligned with maker goals such as creativity, problem-solving, and collaboration.

Conclusion

Solving problems and seeking explanations is an inherently complex process marked by uncertainty, including wrong turns, roadblocks, and trial of alternatives, which demands imagination and creativity. As teachers, we should design instruction that embraces these challenges, helping students build resilience and develop cognitive tools that support creativity and problem-solving. By providing stimulating and thoughtfully designed learning experiences, we can cultivate students' capacities to think and learn. Students should engage deeply with scientific principles and concepts while actively participating in the practices of science. Moreover, nurturing students' imaginations is vital, as creativity is a powerful driver in science learning. Integrating STEAM education, maker education, and DT into our lessons offers tremendous potential to foster emerging 21st-century competencies and skills, equipping students to thrive amid future economic and societal challenges.

In this project, a STEAM resource and instructional package is tailored to support busy science teachers facing limited planning time for such hands-on learning in their classrooms. This package enables teachers to deliver active science learning experiences without significantly increasing their preparation demands or adding to their already heavy workloads. It strikes a balance between ready-to-use, practical lesson ideas and the flexibility needed for adaptation to diverse classroom contexts.

Funds of knowledge refer to the practices and understandings children develop through their lived experiences (Moll et al., 1992). In research conducted by Tan et al. (2018), it was found that deliberately utilising children's funds of knowledge, which are deeply connected to their existing social networks, played a key role in supporting their ongoing and

meaningful engagement in making activities. This finding suggests that community ethnography offers a constructive way to foster robust, purposeful making while also leveraging and honouring children's everyday experiences and the knowledge of their communities (Tan et al., 2018).

Moving forward, I would be keen to also explore the implementation and meaningful engagement of students in prolonged, complex projects in place of the one-off maker-centred lesson activities. This could include teaching students the ethnographic methods in constructing surveys, conducting interviews to gather community insights on problems that matter. Thus, community ethnography offers a promising avenue for leveraging students' funds of knowledge to enhance engagement in STEAM and maker education.

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Appendices

Appendix A— Definition of STEAM Education

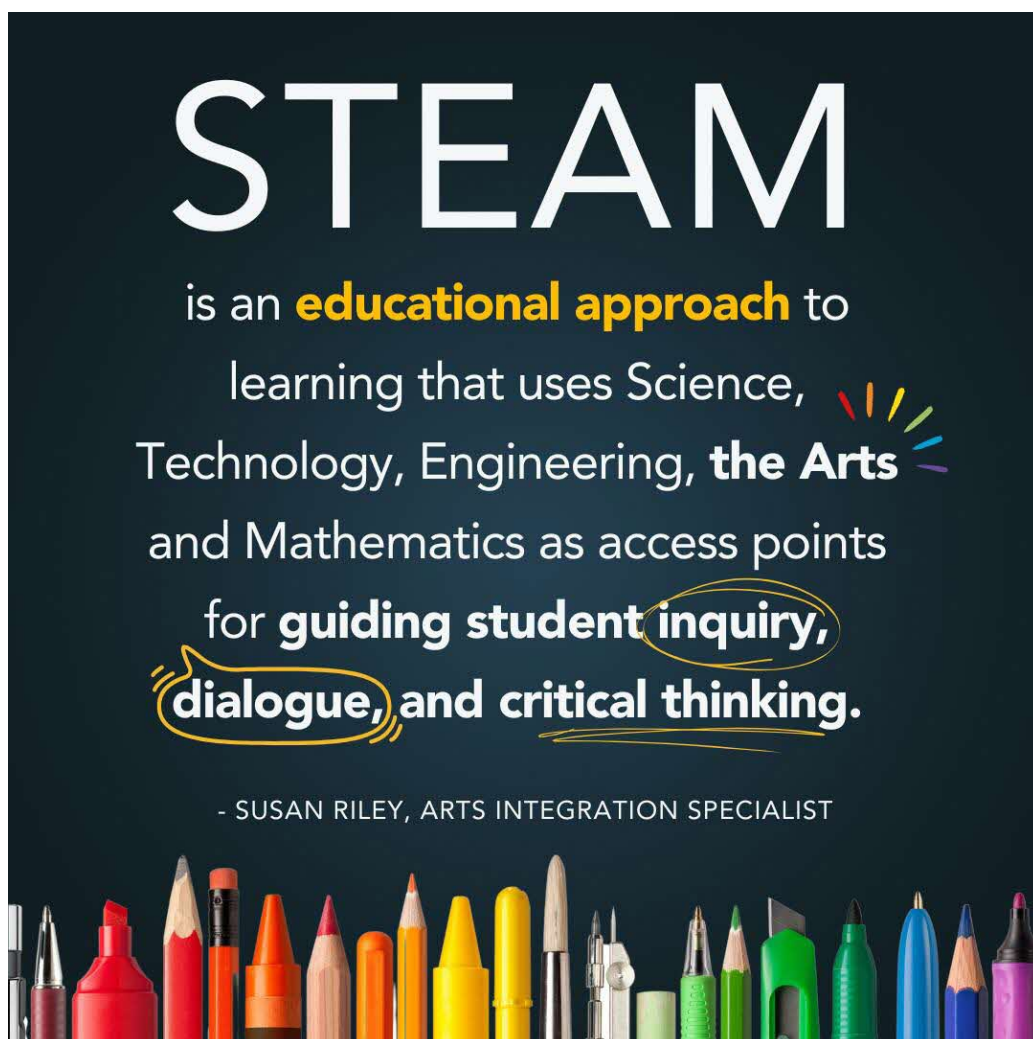
STEAM EDUCATION

STEAM stands for Science, Technology, Engineering, Arts and Mathematics.

“It is a way to take the benefits of STEM and complete the package by integrating these principles in and through the arts. STEAM takes STEM to the next level: it allows students to connect their learning in these critical areas together with arts practices, elements, design principles, and standards”.

Source: The Institute for Arts Integration and STEAM

(<https://artsintegration.com/what-is-steam-education-in-k-12-schools/>)



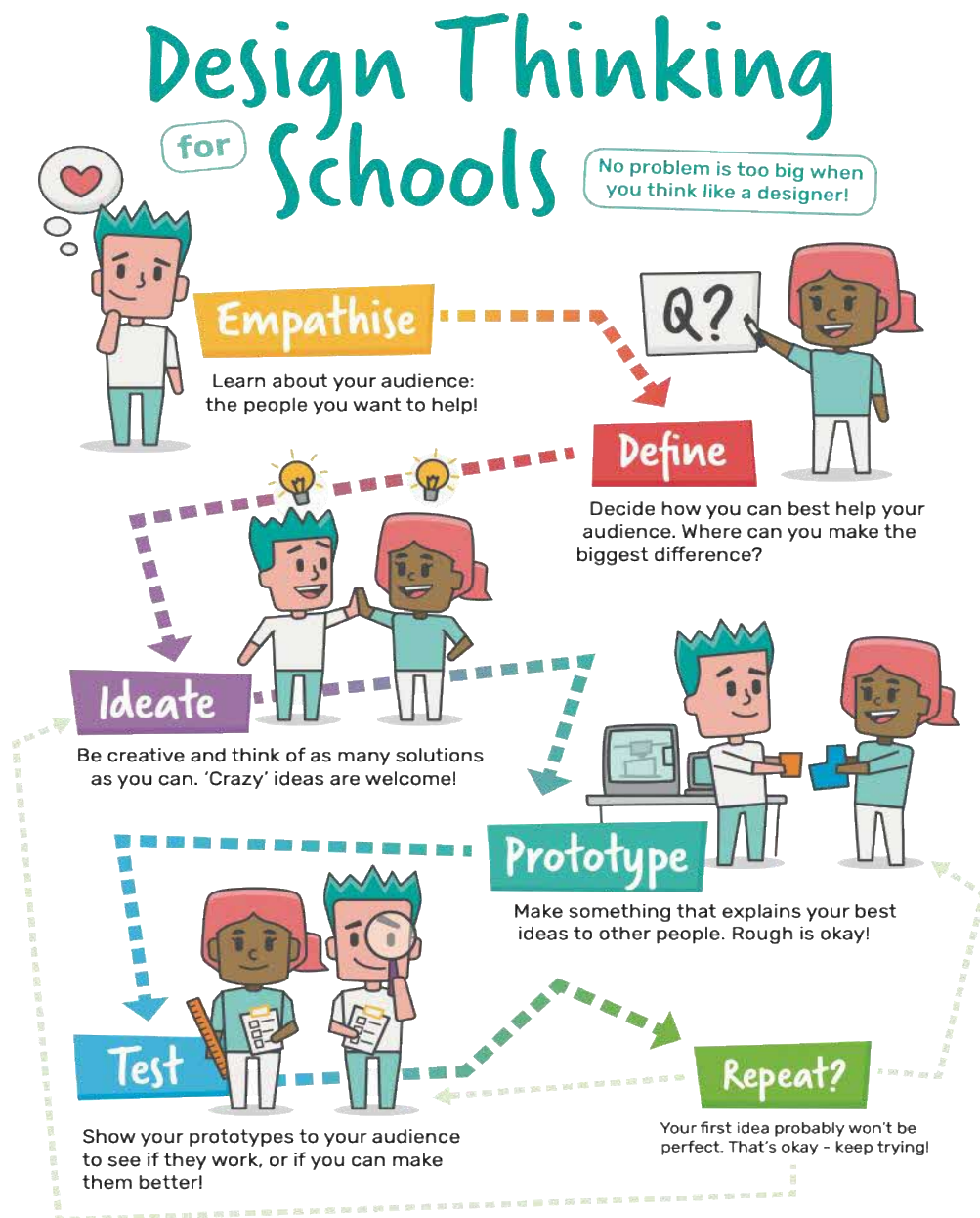
Appendix B— Definition of Design Thinking

What is Design Thinking?

Design thinking is a solutions-based problem-solving approach that encourages students to empathise, generate ideas, and create meaningful outcomes. The infographic below highlights its key stages.

Source: Maker Empire

(<https://www.makersempire.com/design-thinking-for-schools-poster/>)



Source: Personify

(<https://personifycorp.com/blog/design-thinking/>)



Appendix C—Overview of Maker Education

MAKER EDUCATION

A maker mindset, central to maker education, develops the habits and attitudes of curiosity, creativity, and perseverance as students design, build, and refine their ideas. The infographic below highlights the key dispositions that empower learners to think and act like makers.

Source: Kid Museum (<https://kid-museum.org/about/#approach>)



Source: Maker Empire (<https://www.makersempire.com/be-a-maker/>)



BE A
MAKER
(NOT A TAKER)

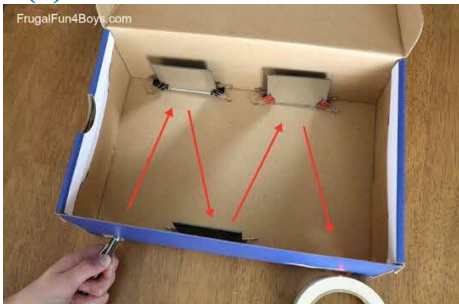
Be creative. Be original. Be curious. Be brave. Be yourself.
Think big. Ask questions. Solve problems. Have fun.
Make new things. Make better things. Make a difference.
You've got this!

Better Learning by DESIGN

MAKERS M EMPIRE

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Appendix D—Two Examples for “Spark and Create— Lesson on a Page”

SPARK AND CREATE- LESSON ON A PAGE (1): Secret Agent Laser Maze		LEVEL& SUBJECT: Primary 4 Science- Light	
LEARNING OUTCOME(S): - Recognise that light travels in straight lines.		PRODUCT EXAMPLE(S): 	
INTERDISCIPLINARY LINK(S): - Language:^[SEP]Writing instructions, describing the design process, storytelling around the secret agent theme, presenting their design to the class. - Art:^[SEP]Creating an aesthetic maze setup. - Physical Education: String laser maze obstacle course		SKILL(S): Collaboration, Communication, Problem-solving, Critical, Adaptive and Inventive thinking SPECIALISED SKILL(S): N.A.	
INTRODUCTION / CONTEXT: Students will role-play as secret agents tasked with protecting a valuable treasure from intruders by constructing a laser maze. Using cardboard, mirrors, and laser torches, students will explore the scientific principle that light travels in a straight line. They will design and build a maze that directs laser beams around corners and across paths, creating a network of light-based “security” to block intruders from reaching the treasure.		REFERENCE(S): https://frugalfun4boys.com/laser-puzzle-boxes/ SAFETY PRECAUTION(S): 1. Handle the mirrors carefully (<i>possible sharp corners</i>) 2. Use the lasers responsibly to ensure they never shine into anyone’s eyes (<i>to only issue lasers after construction</i>)	
MATERIALS (40 students; 8-9 groups): - 9 A4 paper box lids / shoeboxes <i>(To cut 1 hole on each of them for the entrance of laser beam)</i> - 9 pieces of papers for planning - 9 laser pointers - 9 rulers (30-cm) - 45 small square mirrors - 9 boxes of binder clips - 18 pairs of scissors - 9 masking tape rolls - 45 cardboard strips for maze construction - 9 miniature figures (as target / treasure) - Blu tack		DESIGN THINKING STEPS- TO FOCUS ON: ☐ Empathise: what secret agents would need to protect a treasure and why lasers and mirrors might be useful. ☐ Define: How might we create a maze using mirrors to detect and stop intruders from reaching the treasure? ☒ Ideate: Different maze designs, placement of mirrors and resulting laser paths on paper. ☒ Prototype: Build the laser maze using cardboard and mirrors. ☒ Test: Activate laser and test if the light beams follow intended paths; adjust as needed.	
PREPARATION: Place the target/treasure on the box lid for each group. ACTIVITY STEPS: 1. Introduce the context and scenario. Consider showing a short video clip to engage them (E.g., Laser field from movie, Ocean 12) 2. Allow students to work on the following, taking reference from the DT areas to focus on. Provide guidance as needed. Consider setting timers and issue mini challenges in tiers. a. Ideate and sketch the blueprint. b. Prototype and then test. Refine the design as needed. c. Level up the challenge or provide extension if they are done ahead of others. 3. Gallery walk: one member of each group to stay on to demonstrate the completed maze to others.			
DIFFERENTIATION: CHALLENGE FOR DEEPER INQUIRY (C) / EXTENSION FOR BROADER LEARNING (E): - (C) Cut another hole on the cardboard lid: have the light enter and exit the lid (<i>2nd picture under PRODUCT</i>). - (C) Challenge students to create a maze with limited materials. - (C) Add more laser paths for increased complexity (have more than one beam of light to hit the target- would need to have two holes in the lid for entries of 2 beams). - (E) Research real-world applications of lasers and mirrors.		DIFFERENTIATION: MODIFICATION FOR LOW READINESS OR MORE SUPPORT (M) • (M) Vary the number of mirrors that they needed to use. (E.g., Less is easier) • (M) Shifting the target to varied locations	
REFLECTION QUESTIONS: - How did changing the mirror angles affect the laser path? - What challenges did you face in building your maze? - How does understanding light help in real-life situations?		ASSESSMENT/SUCCESS CRITERIA: ☐ All safety precautions are followed throughout the activity. ☐ Light travels continuously without interruption to hit the target precisely. ☐ Maze is constructed within the box, meets the requirements (e.g., number of mirrors) ☐ Demonstrates the ability to show and explain how the maze works to others. ☐ Students effectively collaborate and share responsibilities within their groups.	
REAL-LIFE APPLICATIONS: - Security systems using laser-based alarm detection in museums or banks - Laser communication technology for data transmission - Medical tools employing lasers for surgery and diagnostics - Entertainment and art installations with laser light effects			

SPARK AND CREATE- LESSON ON A PAGE (2): Wayang Kulit / Shadow Puppet Show		LEVEL& SUBJECT: Primary 4 Science- Light & Shadows
LEARNING OUTCOME(S): - Recognise that light travels in straight lines and thus a shadow is formed when light is completely or partially blocked by an object. - Investigate the variables that affect shadows formed. (Shape, size and position of object(s); Distance between light source-object and object-screen)		PRODUCT EXAMPLE(S):  
INTERDISCIPLINARY LINK(S): - Language: Writing scripts for the puppet show and develop storytelling skills as they present their show to the class. - Art: Creative design of puppets - Music: They might compose simple musical patterns or select existing pieces that match the mood or theme of their puppet story.		
INTRODUCTION / CONTEXT: Students will take on the role of storytellers as they create and move shadow puppets to explore how light interacts with objects to form shadows. Using cut-out shapes, a light source and a translucent screen, students will investigate how changing variables such as the distance between the puppet and the light or the puppet and the screen affects the size and sharpness of shadows. Through this hands-on activity, students will deepen their understanding of the properties of light and the shadow formation.		
MATERIALS (40 students; 8-9 groups): - for class presentation, 1 big screen (<i>upcycle a big box with tracing paper</i>) - for each table group to rehearse, 9 small screens (<i>upcycle A3 sized boxes with tracing paper</i>) - 9 pieces of papers for planning - 45 pieces of cardboard paper for puppet shapes - 45 sticks/straws to hold puppets - 9 torchlights - 18 pairs of scissors - 9 masking tape rolls - 9 glue sticks		DESIGN THINKING STEPS- TO FOCUS ON: ☐ Empathise: Consider the needs of their audience—what would capture their classmates’ interest and imagination? ☐ Define: How might we create an engaging shadow puppet show that teaches others about how shadow size changes? ☒ Ideate: Brainstorm a variety of story ideas, puppet shapes, and ways to create shadow effects. ☒ Prototype: Draw and cut out the puppets. Rehearse their stories, experimenting with puppet movement and light positions. ☒ Test: After presentation, students reflect on what worked, what could be improved, and adjust for future presentations.
DIFFERENTIATION: CHALLENGE FOR DEEPER INQUIRY (C) / EXTENSION FOR BROADER LEARNING (E): • (C) Introduce transparent coloured materials to create coloured shadows and explore how light passes through different materials. • (C) Student measure shadow lengths at multiple distances and create a graph to analyse the relationship. • (C) & (E) Student write and perform an original shadow puppet story with their own characters (<i>with integrated scientific explanations of how shadow change.</i>) • (E) Combine with simple music composition for the shadow puppet show that incorporates timing cues and mood, enhancing cross-disciplinary creativity		PREPARATION: Use a fixed distance between light source and screen and only vary puppets’ positions. ACTIVITY STEPS 1. Introduce the context and scenario- making link to local cultural heritage (E.g., Asian Civilisation Museum’s Wayang Kulit video) 2. Allow students to work on the following, taking reference from the DT areas to focus on. Provide guidance as needed. a. Ideate and draft the storyline and determine the number of characters. (<u>Criteria: the story must involve change(s) in size of a character or object.</u>) b. Prototype and then test (make the puppets and test out how to vary shadow size). Adjust as needed. c. Level up the challenge or provide extension if they are done ahead of others. 3. Groups will take turns to present.
REFLECTION QUESTIONS: • Why do you think shadows get bigger or smaller with distance changes? • If you could change something in your puppet or setup, what would you try next time?		DIFFERENTIATION: MODIFICATION FOR LOW READINESS OR MORE SUPPORT (M) • (M) Provide pre-cut puppet shapes instead of having students cut their own to focus more on the light and shadow investigation. ○ Or go old-school with hand shadows. • (M) Provide ready stories / divide up the Gruffalo story into segments so that each group focus on just one short presentation.
REAL-LIFE APPLICATIONS: • Understanding how light creates shadows enhances the design of stage lighting and visual effects. • Manipulating shadows and light is crucial for creating mood, depth, and focus on images and movies. • Knowledge of light and shadows helps architects plan natural lighting, shade structures, and optimize energy use in buildings.		ASSESSMENT/SUCCESS CRITERIA: ☐ All safety precautions are followed throughout the activity. ☐ Demonstrates skill in manipulating the puppet and light source to change shadow size and shape. ☐ Students use puppet movement and size variation intentionally to enhance the storytelling ☐ Demonstrates the ability to show and tell the story clearly to others. ☐ Students effectively collaborate and share responsibilities within their groups.

Appendix E—Idea Sprint (Work in Progress)

Idea Sprint: Collection of Science Activities (within curriculum)

Level/Topic	Concept	Task	Prototype
Lower-Middle Primary / General Science	- Encourages observation and classification skills. - Supports science vocabulary and visual learning. Possible themes: - Shapes of clouds - Seeds, fruits or small insects - Leaves shapes and venation. - Flowers shapes and structures. - Footprint left by animals.	Students will design and construct a handheld “nature viewer” using cardboard as the main material. They will paste or draw plant and animal images around the frame’s window or lens area. There must be two different topics, one on each side. As they explore outdoors, students can look through their frame to help compare and identify real-life specimens with their guide images.	

P3 / Diversity of plants Plant system P6/ Adaptation of plants	1. A plant is made up of leaves, stem and roots. 2. Plants have unique shapes, textures, and structures due to adaptations to survive better.	Students will create unique sculptural artworks by assembling plants, dried branches, and leaves. Through this maker activity, students will explore the natural world as an artistic resource, experimenting with design and structure while learning how to responsibly gather and utilise found materials.	
P5 / Plant Reproduction	1. Flowers have adaptations and parts which helps to ensure pollination take place.	Students will design and build a three-dimensional model of a flower or pollinator. They will then simulate the process of pollination by transferring “pollen” (such as coloured powder or small beads) between the flower models.	

P6/ Living together	1. A community includes only the populations of different species living and interacting in a particular area 2. Biodiversity. Specific adaptations allow organisms to thrive in environments within a habitat and community.	Students design and construct bracelets or necklaces where each bead represents an organism in a biome. The bead colours and sequence should show relationships like food chains or energy flow. The bracelet should be accompanied with a card explaining the meaning of their design.	 
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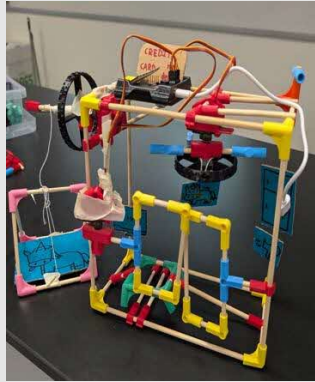
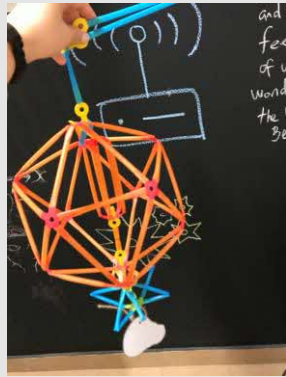
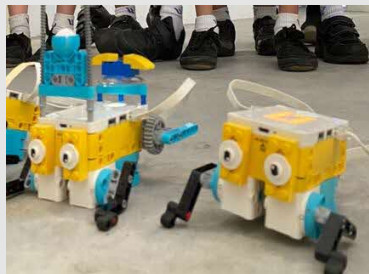
P6/ Living together and Adaptations	1. Food Webs and Predator-Prey Relationships 2. Adaptations and Animal Identification: Through dissecting explore owl adaptations for hunting (like swallowing prey whole) and practice classification skills by matching bones to specific animals.	Students will dissect owl pellets to uncover and identify the bones of animals the owl has consumed. Using the recovered bones, students can reconstruct the possible appearance and structure of the prey. By examining what the owl has eaten, they explore food webs, predator-prey relationships, and how energy moves through an ecosystem.	
P6 / Interdisciplinary Project	1. Heat conductivity of materials; Thermal Insulation and Heat Transfer 2. Sustainability and Environmental Impact	Students will design and prototype a thermal cup that keeps hot drinks hot and cold drinks cold, tailored for a coffee shop setting. Throughout the project, students will integrate principles of thermal insulation, sustainability, and entrepreneurship to create an innovative product with real-world impact.	They will develop a business plan that includes ways to minimise the product's carbon footprint by selecting sustainable materials and processes. The plan will also feature marketing strategies aimed at attracting environmentally conscious customers. 

Appendix F—Maker Playground (Work in Progress)

Maker Playground: Collection of Activities (requiring tools)

Theme	Task / Learning goal(s)	Tools Needed	Prototype
Open theme <i>Recommended for: rapid ideation and visual creativity through digital art creation.</i> A theme or message which student wants to communicate (e.g., environmental awareness, a book summary, a social cause, or a creative story)	Student will use AI image generation tools to create an original image that visually represents their chosen theme, message, or idea. This image will be the centrepiece of a poster, flyer, or communication piece. Learning Goals: • Practice creativity in writing descriptive prompts. • Explore the use of AI tools for authentic visual creation. • Develop skills in visual communication and design. • Reflect on how images and text combine to convey meaning.	Computer device - Laptops - iPads Canva AI Image Generators: Includes Magic Media™ and Dream Lab-powered tools for text-to-image generation with various art styles Padlet includes an AI image generation tool called “I Can’t Draw” that allows users to create images simply by typing a descriptive text prompt. NightCafe features an easy-to-use interface with limited free use and parental controls. Its focus on art styles and safety filters makes it accessible for classroom settings or home use.	AI generated image. Example: • Prompt- Create an image to promote recycling. Show a witch recycling and Santa Claus at the side sorting trash. Image created with Padlet  Image created with Night Café 

Open theme <i>Recommended for: hands-on, tactile experience for creating 3D physical forms freehand.</i> A theme or concept which student want to express with a 3D creation (e.g., an animal, a landmark, a tool, a character from a story, or an abstract art piece).	Students will use a 3D printing pen to design and create a three-dimensional object that represents a theme, idea, or story of their choice. This can be a model, sculpture, functional item, or artistic creation. Learning Goals: • Develop spatial thinking and manual dexterity through 3D building. • Explore open-ended creativity by designing objects meaningful. • Practice problem-solving and planning skills in translating ideas into 3D forms. • Understand the basics of additive manufacturing and prototyping.	3D Printing Pen - 3Doodler Start+ - 3Doodler Flow - Mynt3d Junior 2 By holding the pen like a regular pen and drawing in any direction — including straight up into the air — one can create 3D art, figurines, game pieces, or basic plastic. You may even be able to repair broken objects made from ABS plastic by using a 3D pen to “weld” the pieces back together with melted ABS. OR 1. Sketch your design on paper to plan your 3D shape and structure. 2. Use the 3D printing pen to trace and build your object layer by layer, experimenting with shapes, colours, and textures. 3. Join the pieces together to make a 3D structure, as necessary.	3D models and prototypes. Example: 1. Calendar in cubes  2. Corner bookmark  3. Dog 
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Open theme <i>Recommended for: systematic building with modular components. These kits focus on structural stability, mechanical function, and collaborative construction.</i> A theme or concept which students want to bring to life through hands-on construction and design using modular building kits. (e.g., a vehicle, a building, an animal, a mechanical device, or an abstract model).	Students will use construction kits to design and build a three-dimensional model or structure that represents an idea, theme, or solves a problem of their choice. Learning Goals: • Foster creativity, engineering design thinking, and problem-solving skills. • Develop spatial reasoning and fine motor skills through hands-on building. • Encourage teamwork, communication, and iterative design practices. • Connect physical construction to abstract concepts and real-world applications.	Construction kits: - Stickem - Strawbees - LEGO (LEGO SPIKE comes with sensors, motors, coding kits) 1. Plan and design by sketching or discussing the key features and functions the model should have. 2. Construct your model, experimenting with shapes, balance, and stability. 3. Test your model's functionality or durability if applicable and make improvements.	3D models and prototypes. 1. Stickem  2. Strawbees  3. LEGO 
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Sustainability - upcycling of clothes A theme or concept which students want to explore through the practice of thrifting and creatively upcycling clothes. This project combines sustainability, crafting, and creative expression to create useful, unique accessories. (e.g., transforming old garments into new fashion pieces, accessories, or functional items).	Students will transform thrifted or old clothes - into personalized tote bags - or adding their own designs using hot pressing techniques. - Applying tie-dyeing techniques to colour them - Making soft toys or accessories out of them Learning Goals: • Encourage creative design and practical crafting skills such as sewing and heat transfer. • Promote environmental awareness through recycling and upcycling materials. • Support hands-on experience in transforming everyday items into functional art. • Foster reflection on personal expression and sustainable living.	Resources: - Sewing machines (or hand-held ones) - Needle and thread - Hot-Pressing Machine - Fabric dyes, rubber bands, pail of water and Ziplock bag 1. Source an old shirt from thrift stores, closet, or donations that one wants to repurpose. 2. Plan and create a design or message that reflects the theme or style (e.g., environmental consciousness, a favourite slogan, artistic patterns, or personal symbols). 3. Cut, sew or tie-dye the shirt to form a tote bag shape, using simple stitching or no-sew methods if preferred. Apply your design onto the fabric using heat transfer, hot pressing, or iron-on methods, following safety and technique guidelines.	Prototypes: 1. Upcycled Tee-shirts  2. Tie-dyed shirts  3. Thrifting & tie-dye  4. Soft toys 
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Appendix G—Feedback and Reflection Form for Teachers (Google Form)**Feedback and Reflection form (for teachers)**

This google form helps teachers provide feedback and reflections after using the instructional and resource package. It gathers insights on how effectively the materials supported teaching and learning, what worked well, challenges encountered, and suggestions for improvement to guide future revisions and implementation.



QR code for accessing the form



Reflection for use of MakerSprint STEAM resource package

Thank you for taking the time to provide your valuable feedback on the instructional resource package. Your insights will help us improve the resources and better support your teaching practice.

Your responses will be completely anonymous to encourage honest and open reflections. Please answer the questions thoughtfully.

If you have any further comments or questions, please feel free to reach out.

Thank you.

Best regards,
Huoy Fen

*** Indicates required question**

Level of effort you put in **to implement** the maker-centred lesson. *

Linear scale (1–5):

1 = Not much effort.

5 = Too much effort. Will not do it again.

	1	2	3	4	5	
Not much effort.	☐	☐	☐	☐	☐	Too much effort and will not do it again.

Approximately how much **time did the package save you** when planning your maker-centred lesson? *

Linear scale (1-5):

1 = Spent even more time with planning after using the package (perhaps it confused you)

3 = It did not make a difference.

5 = I saved plenty of time.

	1	2	3	4	5	
Spent even more time with planning after using the package (perhaps it confused you)	☐	☐	☐	☐	☐	I saved plenty of time.

How useful was the resource package for supporting your **teaching goals**? *

Linear scale (1–5):

1 = Not useful

5 = Very useful

	1	2	3	4	5	
Not useful	☐	☐	☐	☐	☐	Very useful

What changes or additions would make "**Lesson on a Page**" more helpful? *

Your answer _____

What changes or additions would make "**Idea Sprint**" more helpful? *

Your answer _____

What changes or additions would make "**Maker Playground**" more helpful? *

Your answer _____

What changes or additions would make **the entire resource package** more helpful? *

Your answer

What **new ideas or practices** did you try as a result of using it?

Your answer

Any other comments or suggestions?

Your answer

Submit

Clear form

Never submit passwords through Google Forms.